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ARIA-NBV: Target-Conditioned, Quality-Driven Next-Best-View Planning

ARIA-NBV: Zielkonditionierte, qualitaetsgetriebene Next-Best-View-Planung

Master's Thesis Proposal / Proposal zur Masterarbeit

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Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the proposal text. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

Contents

1 Motivation

Active reconstruction is a coupled sensing and inference problem: the reconstruction error after a fixed budget depends as much on the selected viewpoints as on the downstream surface estimator. Classical active perception therefore treats sensing actions as part of perception itself, and view-planning surveys formalize the dominant generate-score-select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. ARIA-NBV adopts that loop for egocentric indoor data, but asks a narrower scientific question: can finite candidate views be scored and planned by reconstruction-quality improvement rather than by coverage or uncertainty proxies alone?

The key empirical precedent is VIN-NBV, which replaces pure coverage with Relative Reconstruction Improvement (RRI), an oracle label computed from the reduction in Chamfer-style reconstruction error after adding a query view [Fra+25]. This is the right axis for ARIA-NBV because target indoor surfaces can remain poorly reconstructed even when many voxels or pixels are already covered. The thesis extends this idea from object-centric RGB-D benchmarks to the Project Aria / ASE ecosystem, where calibrated egocentric streams, trajectories, semi-dense points, predicted object boxes, and a mesh-supervised subset support controlled oracle labels [Eng+23, Met25, Str+24a].

The current gap is not a lack of possible algorithms; it is the lack of a trustworthy target-conditioned, multi-step evidence chain. GenNBV and Hestia show that continuous 5-DoF and hierarchical NBV policies are plausible when an online simulator and coverage reward are mature [Che+24, Lu+26]. Radiance-field and Gaussian-splatting papers show that view utility can be decomposed into uncertainty, Fisher information, semantics, dynamics, object identity, or downstream task error [BLL25, Jeo+26, JLD24, LJD25, Pan+22, Str+24b]. ARIA-NBV should use those papers to sharpen the model, not to replace its objective: the thesis utility remains scene and target RRI, with validity and cost reported as separate constraints.

Thesis position

The thesis contribution is a target-conditioned, quality-driven finite-candidate NBV study on ASE/EFM. The hard result is a candidate-query Q_H model that predicts bounded cumulative target RRI over valid candidates and improves on one-step greedy/model scoring under the same acquisition budget. Continuous actor-critic control, Gymnasium/SB3, Habitat/Isaac, SceneScript-style global memory, and real-device guidance are bridge or future work unless this finite-candidate result is already stable.

2 Problem and Research Contract

ARIA-NBV is a target-conditioned, finite-candidate NBV problem in the Project Aria / ASE / EFM observation regime. The thesis is deliberately not a first-order continuous-control claim: it tests whether bounded planning over a finite valid candidate table improves target reconstruction quality beyond myopic view selection.

The actor-visible state at decision step t is

$$\boldsymbol{s}_t^{\mathrm{obs}} = \left(\boldsymbol{P}_t^{\mathrm{semi}}, \boldsymbol{F}_t^{\mathrm{EVL}}, \boldsymbol{O}_t^{\mathrm{pred}}, \boldsymbol{h}_t, b_t\right),$$

where $\boldsymbol{P}_t^{\mathrm{semi}}$ is the accumulated semi-dense or fused point evidence, $\boldsymbol{F}_t^{\mathrm{EVL}}$ is the frozen local EFM/EVL evidence field, $\boldsymbol{O}_t^{\mathrm{pred}}$ are observed or predicted target hypotheses, $\boldsymbol{h}_t$ is selected-view history, and b_t is remaining budget. The oracle state augments this with ASE GT assets:

$$\boldsymbol{s}_t^{\mathrm{oracle}} = \left(\boldsymbol{s}_t^{\mathrm{obs}}, \boldsymbol{M}_{\mathrm{GT}}, \left\{\boldsymbol{M}_e^{\mathrm{GT}}\right\}_{e\in\mathcal{E}}, \left\{\boldsymbol{P}_{t,i}^{\mathrm{cand}}\right\}_{i=1}^{N_q}\right).$$

GT meshes, GT OBBs, GT crops, GT semantic labels, and all-candidate GT renders are label/evaluation assets only. They are not V1 actor inputs.

The planner state is

$$\boldsymbol{s}_t^{\mathrm{cf0}} = (\boldsymbol{F}_0^{\mathrm{EVL}}, \boldsymbol{P}_t, \boldsymbol{z}_e, \boldsymbol{h}_t, b_t, \boldsymbol{Q}_t, \boldsymbol{m}_t, \boldsymbol{\rho}_t),$$

where $\boldsymbol{Q}_t = \{\boldsymbol{q}_{t,i}\}_{i=1}^{N_q}$ is the finite candidate table, $m_{t,i} \in \{0,1\}$ is a hard validity mask, and $\rho_{t,i}$ is an invalid reason. The admissible actions are candidate indices:

$$\mathcal{A}_t = \{i \in \{1, \ldots, N_q\} : m_{t,i} = 1\}.$$

Invalidity is a constraint, not low utility. Masks apply before argmax, softmax, loss targets, and bootstrap maximization.

The V1 target protocol is OBS-SEL / PRED-Q / GT-EVAL. Target selection and model input use an actor-visible descriptor

$$\boldsymbol{z}_e = \varphi\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{p}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \boldsymbol{T}_e^{\text{rel}}\Big),$$

covering predicted/observed OBB geometry, class, confidence, projected area, semi-dense support, EVL support, and relative pose. GT crops $\boldsymbol{M}_e$ are used only after matching $\boldsymbol{z}_e$ to GT by class compatibility, OBB overlap, and support. Unmatched or ambiguous targets are reported as target-invalid cases.

For target e, the implemented oracle error is the point-mesh accuracy plus mesh-to-point completeness diagnostic:

$$\Delta_t^e = \mathcal{A}_t^e + \mathcal{C}_t^e.$$

This is the `pm_acc_*` / `pm_comp_*` distance used by `aria_nbv`, not generic point-cloud Chamfer distance. If valid action $a_t = i$ selects $\boldsymbol{q}_{t,i}$, then

$$\boldsymbol{P}_{t+1} = \boldsymbol{P}_t \cup \boldsymbol{P}_{t,i}^{\text{cand}}, \quad r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_t^e + \varepsilon}.$$

The value-learning return and endpoint metric stay separate:

$$G_t^H = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e, \quad J_{e,H} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}.$$

The log-gain companion $L_{e,H} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$ is only an ablation.

Main question

Can a target-conditioned candidate-query Transformer $Q_{H,\theta}\big(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}\big)$ choose valid candidate views whose oracle-evaluated cumulative target RRI beats one-step greedy/model scoring under equal acquisition and candidate budgets?

The hard quantitative core is observed target selection, target RRI labels, a target-conditioned one-step scorer, replayable oracle rollouts, and mandatory Q_H. Online discrete Q_H, IQL, actor-critic control, external simulators, 3DGS control, SceneScript, VLM planning, and real-device guidance are bridge or future-work surfaces unless M5 evidence justifies escalation.

3 Related Work and Positioning

The local paper manifest and shared bibliography were treated as a design corpus, not as a citation list to pad the proposal. The scientific pattern that emerges is consistent: older active-perception and view-planning work motivates action-conditioned sensing; modern NBV work splits between continuous coverage-driven policies, quality-driven finite candidate ranking, projection shortlists, and radiance-field uncertainty; offline RL contributes support and overestimation warnings for finite-candidate value learning.

Corpus family	Main technical signal	ARIA-NBV use
Active perception and classical view planning [AWB88, Baj88, Ban+00, SRR03]	Viewpoint choice changes what constraints are observable; candidate generation and feasibility are first-class parts of perception.	Keep a finite candidate table, explicit validity, and acquisition budget instead of hiding planning behind a black-box policy.
Receding, projection, and finite-candidate NBV [Bat+22, Bir+16, Fra+25, Jia+25]	Evaluator cost, horizon, branch factor, and reconstruction-quality labels determine whether deeper search is meaningful.	Use mesh-supervised RRI as the utility label; use projection/frontier ideas only as proposal or diagnostic channels.
Egocentric substrate [Eng+23, Met25, Str+24a]	Project Aria supplies calibrated egocentric streams; ASE adds synthetic scale, GT meshes, OBBs, trajectories, and EVL-style 3D state.	Use predicted/observed OBB and EVL support as actor-visible target state; keep GT meshes/boxes for labels and evaluation only.
Continuous and hierarchical learned NBV [Che+24, Lu+26, Sch+17]	Coverage-driven PPO and hierarchical look-at-then-fly policies become powerful when simulator state, reward, and dynamics are mature.	Use as bridge design and baseline context after finite-candidate RRI evidence shows headroom.
Radiance-field / 3DGS active selection [BLL25, Jeo+26, JLD24, Ker+23, LJD25, Mil+20, Pan+22, Str+24b]	Uncertainty, Fisher information, depth, semantic, dynamic, object-specific, and downstream-task utilities can be logged separately.	Adopt utility-channel separation and target focus; do not replace target RRI with 3DGS uncertainty before calibration.
Offline value learning and sequence planning [Che+21, FMP19, Haa+17, Haa+18, HGS15, JLL21, KHW19, KNL21, Kum+20, Mni+13]	Replay, beam decoding, stochastic sampling without replacement, max-over-action overestimation, and offline support mismatch are the relevant hazards.	Train masked fitted Double-Q Q_H first; keep IQL, sequence decoding, soft/energy policies, and actor-critic control as gated ablations.
Structured semantic scene representations [Ave+24, Xie+25]	Autoregressive scene languages and editable layouts support semantic/global memory and human correction.	Use as future global planner context only after observed target contracts and target RRI are stable.

Table 1: Source-backed literature positioning for the ARIA-NBV thesis scope.

The thesis therefore does not claim novelty through a larger continuous action space. Its scientific contribution is the controlled replacement of proxy utility by target-conditioned reconstruction-quality utility under a finite candidate contract. The closest methodological lineage is

$$\mathcal{U}_{\text{cov/unc}} \to \hat{r}_t^e(i) \to r_t^e \to G_t^{(H)} \to Q_{H,\theta}.$$

This lineage makes three negative claims explicit. First, coverage is a useful diagnostic but not the thesis objective. Second, GT meshes and GT target boxes are oracle assets, not actor inputs. Third, offline/continuous RL methods are not safe shortcuts until rollout support, masks, and evaluation are reliable.

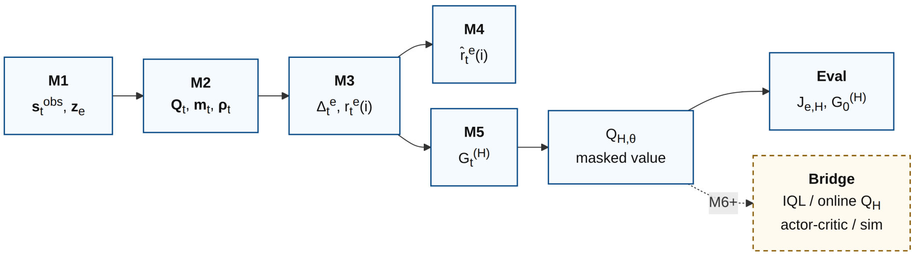

Figure 1: Thesis evidence chain. Core gates move from $\boldsymbol{s}_t^{\text{obs}}$ and $\boldsymbol{Q}_t, \boldsymbol{m}_t$ to r_t^e, $G_t^{(H)}$, and $Q_{H,\theta}$; the dashed node is post-Q_H bridge work.

4 Objectives and Hypotheses

The thesis objective is a reproducible target-aware NBV stack whose selected views improve target reconstruction quality under a fixed budget. The primary hypothesis is policy-level: after oracle re-evaluation, $Q_{H,\theta}$ should achieve higher $J_{e,H}$ and G_0^H than myopic learned or greedy one-step selection on identical roots, candidates, masks, and budgets.

O1: Trustworthy oracle labels. M1 must validate offline-store indexing, splits, frames, CW90 display handling, candidate-label order, rendered depth, backprojection, invalidity reasons, and Rerun inspection before any target or rollout scale-up.

O2: Leakage-safe target utility. V1 uses observed/predicted target descriptors for selection and model input. GT crops and GT boxes are restricted to labels, upper bounds, and evaluation. Scene RRI, target RRI, and acquisition cost are reported separately.

O3: Myopic control. The target-conditioned one-step scorer predicts target RRI over the same candidate table as Q_H. It is evaluated by rank correlation, top-k oracle hit rate, calibration, target visibility, invalid fraction, and failure groups.

O4: Replayable multi-step data. Rollouts include random-valid, oracle-greedy/lookahead, and oracle-scored temperature-softmax traces with seeds, policy ids, masks, reasons, target identity, and candidate provenance. Gumbel-Top-k is later diversity evidence after deterministic lookahead is trusted.

O5: Mandatory finite-candidate Q_H. The planner emits one masked value per candidate:

$$\boldsymbol{u}_{t,i} = \mathrm{Tr}_{\theta(\boldsymbol{S}_t, \boldsymbol{T}_e, \boldsymbol{H}_t, \boldsymbol{C}_{t,i})}, \quad Q_{H,\theta}\big(\boldsymbol{s}_t^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}\big) = \boldsymbol{w}_Q^\top \boldsymbol{u}_{t,i},$$

$$a_t^\theta = \operatorname*{argmax}_{i \in \mathcal{A}_t} Q_{H,\theta}\big(\boldsymbol{s}_t^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}\big).$$

Success is measured by oracle-rescored selected actions, not by predicted values. If Q_H does not beat one-step controls, the thesis reports the failing gate rather than replacing the core claim with unvalidated continuous RL.

Claim	Metric	Evidence rule
Target utility	$J_{e,H}$, G_t^H, scene RRI, cost	Endpoint gain and cumulative return are separate; budgets use quality-cost curves.
Target input safety	selector rank, match score, support, leakage checks	V1 actor input is observed/predicted; GT produces labels/evaluation only.
Candidate space	valid fraction, reasons, target visibility, provenance	Hard masks and strategy provenance are present before scorer or Q_H training.
Planning gain	oracle-evaluated target RRI under equal budget	Q_H is compared to random-valid, one-step greedy, learned one-step scoring, and oracle lookahead.
Scale	scenes, snippets, targets, trajectories, seeds, gaps	Full ASE GT-mesh coverage or an exact held-out subset report.

Table 2: Objective-to-evidence matrix.

5 Method

The method is a sequence of empirical gates. Each gate either produces a trusted artifact for the next gate or records a blocker.

5.1 M1/M2: Geometry and One-Step Baseline

First, candidate poses, rig poses, rendered depths, backprojected point clouds, semidense support, EVL fields, and Rerun diagnostics must agree in frame and indexing. The oracle labels use the implemented point-mesh diagnostics

$$\Delta_t = \mathcal{A}_t + \mathcal{C}_t, \quad \Delta_t^e = \mathcal{A}_t^e + \mathcal{C}_t^e, \quad r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_t^e + \varepsilon}.$$

The one-step VIN-style scorer is then trained as the myopic comparator. Its purpose is to test whether actor-visible state and candidate features predict oracle RRI well enough to justify planning.

5.2 M3/M4: Target Contract and Target Scoring

The selector returns a small top-K set of observed/predicted targets. V1 refuses GT OBB input; the labeler matches selected targets to GT crops by class compatibility, OBB overlap, visibility/support, projected area, and semidense/EVL support. Ambiguous matches are excluded and counted.

The target-conditioned scorer receives the same finite candidate set as Q_H. Its ablations are bounded to target descriptor choice, surface input, CORAL variant, auxiliary regression, candidate-relative pose encoding, and calibration/stage features.

5.3 Candidate and Replay Contract

Candidate generation is logged as a finite mixture over target-point, radial shell, and forward-rig continuity families. Optional projection/frontier shortlists are diagnostics until calibrated. Every candidate row stores provenance, pose, target/scene support, path increment, validity mask, invalid reason, and oracle labels. Candidate order is not semantic; shuffled-candidate evaluation is required for Q_H.

The minimum replay row is

$$\left(\text{scene}, \text{snippet}, \text{target}, t, \boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{Q}_t, \boldsymbol{m}_t, \boldsymbol{\rho}_t, a_t, r_t^e, \boldsymbol{s}_{t+1}^{\text{cf0}}, \boldsymbol{Q}_{t+1}, \boldsymbol{m}_{t+1}, \text{policy}, \text{seed}\right).$$

This is enough to reproduce the mask, value target, and oracle re-evaluation.

5.4 M5: Rollouts and Q_H

Generate deterministic oracle greedy and bounded oracle lookahead first:

$$a_t^{\text{greedy}} = \underset{i \in \mathcal{A}_t}{\operatorname{argmax}}\, r_t^{e(i)},$$

$$a_t^{\text{look}} = \pi_0 \left(\underset{a_{t:t+H-1}}{\operatorname{argmax}} \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e \right).$$

Temperature-softmax traces then widen support:

$$P(a_t = i \mid \boldsymbol{s}_t) = \frac{\exp(\beta \ell_{t,i})}{\sum_{j \in \mathcal{A}_t} \exp(\beta \ell_{t,j})}.$$

The candidate-query Transformer consumes scene, target, history, and candidate tokens $\boldsymbol{X}_t = [\boldsymbol{S}_t, \boldsymbol{T}_e, \boldsymbol{H}_t, \{\boldsymbol{C}_{t,i}\}]$ and decodes one value per candidate:

$$\boldsymbol{u}_{t,i} = \mathrm{Tr}_{\theta(\boldsymbol{X}_t)_i}, \quad Q_{H,\theta}(\boldsymbol{s}_t^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}) = \boldsymbol{w}_Q^{\top}\boldsymbol{u}_{t,i}.$$

Invalid candidates are filled with $-\infty$ before argmax, softmax, and bootstrap maximization.

With online parameters θ and target parameters θ^{-}, the first target is masked Double-Q:

$$i^{\star} = \operatorname*{argmax}_{i \in \mathcal{A}_{t+1}} Q_{H,\theta}(\boldsymbol{s}_{t+1}^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t+1,i}),$$

$$y_t = r_t^e + \gamma Q_{H,\theta^{-}}(\boldsymbol{s}_{t+1}^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t+1,i^{\star}}),$$

$$\mathcal{L}_{Q(\theta)} = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a}\big(Q_{H,\theta}(\boldsymbol{s}_t^{\mathrm{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,a}) - y_t\big)^2.$$

CQL, BCQ, IQL, Decision Transformer, soft/energy policies, PPO, and SAC remain comparison references or later ablations [Che+21, FMP19, Haa+17, Haa+18, KNL21, Kum+20, Sch+17].

5.5 Evaluation

All selected actions are oracle-evaluated under the same budget. The main table compares random-valid, one-step oracle greedy, learned one-step target scorer, bounded oracle lookahead, temperature-softmax rollouts, and learned Q_H on scene-level splits. Required columns are $J_{e,H}$, G_0^H, scene RRI, view count, path length, invalid-action rate, runtime, scenes, snippets, targets, trajectories, rollout seeds, transitions, and coverage gaps.

Gate	Artifact	Fail-safe
M1	geometry/oracle contract report	block target/RL scale-up if frames or labels are not trustworthy
M3	V0/V1 target RRI and observed-only selector	report target oracle only on validated subsets if matching is sparse
M4	target-conditioned scorer	freeze as myopic baseline with calibration and failure groups
M5	rollout store, oracle lookahead, Q_H	report the exact blocker if Q_H does not beat myopic controls
M6+	online/IQL/actor-critic/simulator bridge	do not promote bridge work without M5 evidence

Table 3: Method gates and fail-safe interpretation.

6 Schedule and Risk Control

The planned thesis window runs from 29 April 2026 to 30 September 2026. The schedule is organized around evidence gates rather than feature optimism: offline-store trust and oracle correctness precede target-conditioned modeling; target RRI and observed target selection precede Q_H data generation; bridge work begins only after the finite-candidate planning evidence is interpretable.

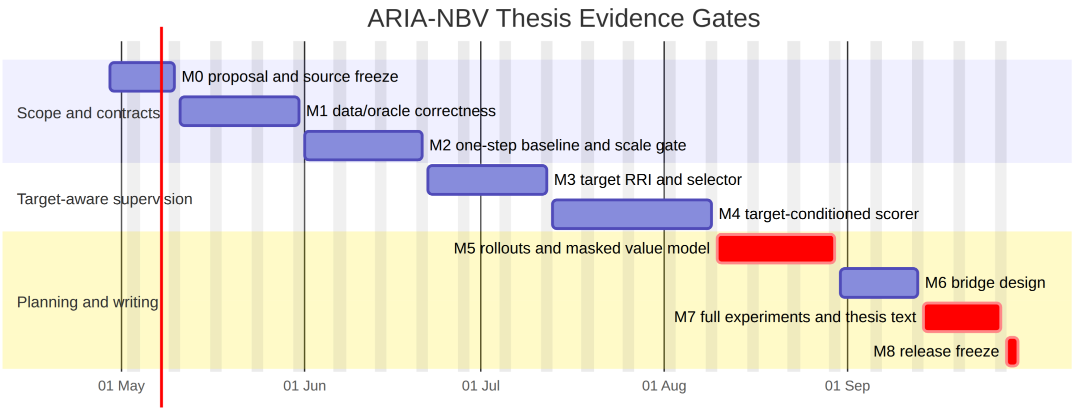

Figure 2: Mermaid Gantt chart for the proposal timeline and evidence gates.

Dates	Milestone	Exit condition
2026-04-29 to 2026-05-10	M0 scope/proposal	Proposal, roadmap, questions, source policy, and hard Q_H boundary are aligned and renderable.
2026-05-11 to 2026-05-31	M1 data/oracle	Offline-store, split, frame/CW90, candidate-label, depth/backprojection, Rerun, and throughput evidence pass or block scale-up.
2026-06-01 to 2026-06-21	M2 one-step baseline	Scene-level VIN baseline, calibration/ranking plots, LRZ sharding plan, and Zarr rollout/Q schema are ready.
2026-06-22 to 2026-07-12	M3 target RRI	V0 GT-OBB target labels and V1 OBS-SEL / PRED-Q / GT-EVAL contract are trusted on a small subset.
2026-07-13 to 2026-08-09	M4 target scorer	Target-conditioned one-step scorer is evaluated against scene-level scoring and oracle target RRI.
2026-08-10 to 2026-08-30	M5 rollouts/Q_H	Random-valid, oracle-greedy/lookahead, and temperature-softmax traces are stable; candidate-query Q_H is trained and oracle-evaluated.
2026-08-31 to 2026-09-13	M6 bridge design	Online discrete Q_H, IQL, actor-critic, hierarchy, and simulator bridge are written as designs or gated ablations.
2026-09-14 to 2026-09-27	M7 experiments/writing	Final tables, figures, failure cases, coverage report, and thesis narrative are frozen.
2026-09-28 to 2026-09-30	M8 release	Configs, smoke checks, demo path, and final PDF artifacts are reproducible.

Table 4: Milestone exit criteria.

The most likely failure mode is not that continuous RL is needed too early; it is that target labels, validity masks, or scale generation are not trusted enough for learned planning. The fallback policy is therefore conservative. If M1 fails, the thesis remains a geometry/oracle contract and one-step scorer study. If M3 target matching is sparse, target RRI remains an oracle diagnostic while scene-level scoring is reported honestly. If Q_H fails to beat myopic controls, the thesis reports whether the limiting factor is candidate support, target observability, reward definition, or model capacity. None of these fallbacks reclassifies the mandatory Q_H attempt as optional.

7 Preliminary Thesis Outline

The thesis will be written as an empirical methods thesis. The chapter order mirrors the validation order so that claims are introduced only after the corresponding contract is established.

Chapter	Purpose	Expected evidence
Introduction	Motivate target-conditioned, quality-driven NBV for egocentric indoor reconstruction and state the finite-candidate thesis claim.	Research questions, contribution boundary, and source-backed scope.
Background	Review active perception, NBV, RRI, ASE/Project Aria, EFM3D/EVL, target-aware 3DGS, and offline value learning.	Literature synthesis with adoption/rejection decisions.
Data and geometry contracts	Describe snippets, calibration, frames, offline stores, candidates, rendered depths, backprojection, masks, and Rerun inspection.	M1 contract report, visual diagnostics, throughput, and known limitations.
Oracle RRI and target RRI	Define scene/target distances, crop matching, invalidity, and label generation.	Label distributions, target crops, target/scene divergence, and failure cases.
Target-conditioned scoring	Present actor-visible target encoding, candidate features, VIN-style model, ordinal/regression losses, and calibration.	Held-out ranking, top-k oracle hit, ablations, calibration, and target-specific failures.
Bounded rollout and Q_H	Compare random-valid, one-step greedy, learned one-step scorer, oracle lookahead, temperature-softmax traces, and candidate-query Q_H.	Cumulative target RRI, endpoint target gain, scene RRI, cost, invalidity, runtime, and rollout visualizations.
Discussion and conclusion	Interpret limits, scale blockers, simulator gaps, semantic/global planning, and real-device bridge paths.	Evidence-gated conclusion and reproducibility package.

Table 5: Preliminary thesis chapter outline.

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction.

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